

Link Failure Prediction and Localization in Cloud-Scale Networks using Supervised Learning

UE24CS352A Machine Learning — Mini-Project Write-up | Team: J Sanjay Varma (PES1UG24CS194) & Kanak Pandey (PES1UG24CS212)

1. Problem statement

Cloud providers carry traffic over optical links, and each end-to-end (E2E) link crosses many fibre spans and inline amplifiers. Fibres that slowly get lossier, amplifiers that lose gain, loose connectors and fibre cuts all reduce the optical signal-to-noise ratio (OSNR) until traffic fails. Following Bakhtiari [1], we predict from routine telemetry whether a link is **healthy** or **not healthy**, so operators can act before customers see an outage. We then **localise** the faulty span and element so a repair can be scheduled.

2. Dataset

The paper's live network data is not public, so we built a physics-based telemetry simulator that matches its format: **20 directional E2E links** (10 routes $\times$ 2 traffic directions), **19 spans** each, and 4 raw metrics per span every **2 minutes** (span loss, target span loss, amplifier gain, target gain). This gives **6000 samples**, split 75/25 into 4500 for training and 1500 for testing, as in the paper. We injected four fault types with known locations: fibre degradation, amplifier degradation, flapping connector and fibre cut. Slow faults ramp up gradually; others are abrupt steps. As in the paper, labels come from OSNR. We compute a generalised OSNR from an ASE + GN-model nonlinear-noise link budget, and a sample is *not healthy* (24.5% of samples) if any of the following holds: OSNR < 17 dB, received power more than 7 dB below nominal, or loss of signal. The telemetry includes measurement noise, about 2% missing cells, whole missing bins, and no loss readings during fibre cuts.

3. Approach

Preprocessing (paper §III). Loss readings missing because of a fibre cut are filled with the expected cut value (target + 40 dB). All other gaps are filled by linear interpolation in time, separately per link (traffic direction) and span. Per-span differences $x_1 = \text{gain} - \text{target gain}$ and $x_2 = \text{target loss} - \text{span loss}$ (Eq. 1) are summed over the 19 spans to give the two E2E features $X1 = \sum x_1$ and $X2 = \sum x_2$ (Eq. 2).

Classifiers (paper §IV), written from scratch in NumPy. (i) *Logistic regression*: minimises the cross-entropy $J(\theta)$ using Newton's method, converging in 11 iterations to $J = 0.0662$. (ii) *GDA*: closed-form maximum-likelihood estimates of ϕ , μ_0 , μ_1 and a shared Σ , which give a linear boundary at $p(y|x) = 0.5$. (iii) *Linear SVM*: minimises the soft-margin hinge loss in the primal with the mini-batch Pegasos sub-gradient solver. Each model was checked against scikit-learn (99.8–100% agreement on test predictions). **Localisation:** the faulty element is the span with the largest positive loss or gain residual. A link is reported as **flapping** when its predicted health toggles at least 4 times in 30 minutes.

4. Implementation overview

The code is a Python package, **linkfail/**, with modules for configuration, the simulator, preprocessing, models, localisation, evaluation, plots and the pipeline. **run_pipeline.py** regenerates all data, models, metrics and figures in about 20 seconds. **app.py** is a Streamlit demo: it injects a fault into any span and shows each model's decision and the localised span live. **predict.py** provides command-line inference, and a 12-test pytest suite checks the models against scikit-learn, the physics and the preprocessing. Robustness was measured with stratified 5-fold CV and with GroupKFold over routes, so the test links are never seen in training.

5. Results

Classifier	Acc.	Prec.	Recall	F1	AUC	5-fold CV	Unseen routes	Paper
Logistic Regression	96.5%	0.927	0.929	0.928	0.995	97.0% $\pm$ 0.7	96.9% $\pm$ 0.8	99%
GDA	88.9%	0.935	0.589	0.722	0.964	89.9% $\pm$ 1.0	89.9% $\pm$ 2.9	97%
Linear SVM	96.6%	0.927	0.935	0.931	0.995	97.0% $\pm$ 0.6	96.9% $\pm$ 0.8	99%
SVM (RBF kernel) *	97.2%	0.948	0.937	0.942	0.996	–	–	–
LR + 38 per-span residuals *	97.4%	0.953	0.940	0.947	0.997	–	–	–

Table 1. Held-out test set (1500 samples). * = extension beyond the paper.

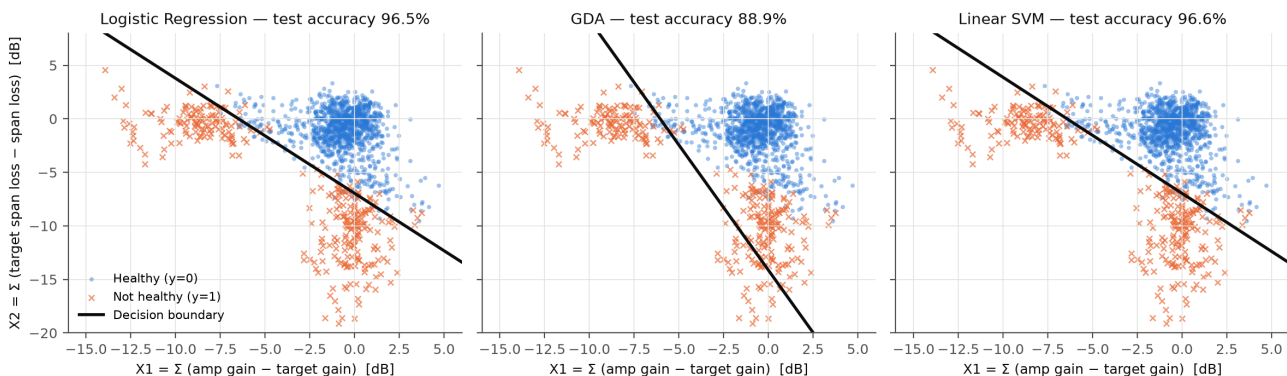


Figure 1. Test data and the learnt linear decision boundaries (compare with paper Fig. 4–5).

Learnt coefficients ($\theta_0, \theta_1, \theta_2$) are LR (-10.44, -1.61, -1.50), GDA (-5.73, -0.95, -0.41) and SVM (-6.18, -0.96, -0.89). For LR and SVM, $\theta_1 \approx \theta_2$, so the rule they learn is "alarm when the net power deficit $X1 + X2$ falls below about -7 ".

dB", which is the physically correct rule. **Localisation** identified the right span and element type for 100.0% of detected failures ($n = 341$). Over every fault-present sample, including minor faults, span accuracy was 91.6%, and a learnt multinomial-LR localiser reached 99.7%. **Flapping** detection found 9 of 9 events, with a 7.1% false-alarm rate. **Early warning**: 18 of 18 degradations that ended in an outage were flagged no later than the outage, and 12 were flagged before it (mean lead time 8 min).

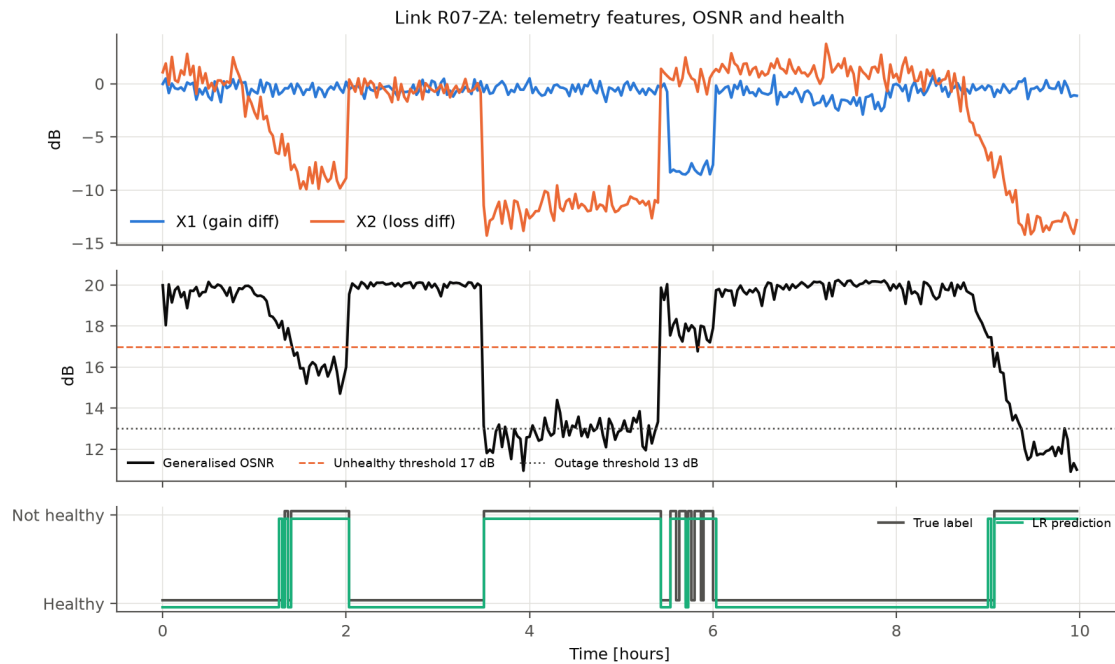


Figure 2. Ten hours of one link: features, physical OSNR, and the true label against the logistic-regression prediction (out-of-fold, link unseen in training).

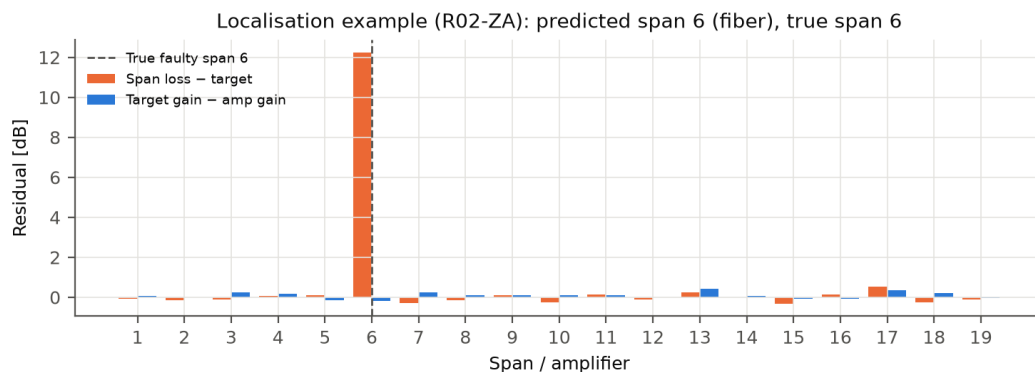


Figure 3. Per-span residuals of a detected failure; the largest residual pinpoints the faulty span and whether the fibre or the amplifier is at fault.

6. Discussion and conclusions

We reproduce the paper's ranking: **logistic regression $\approx$ linear SVM (96.5% / 96.6%) > GDA (88.9%)**. Accuracy is the same on unseen routes, so the models generalise across links. GDA trails because its assumption of one Gaussian per class with a shared covariance does not hold. The not-healthy class has several clusters: gain loss, fibre loss, and fibre cuts at $X2 \approx -40$ dB. These stretch Σ and tilt the boundary, so GDA's recall drops to 0.59. Our absolute accuracy is lower than the paper's 99% because our labels follow OSNR physics. Summing over 19 spans discards where the fault is: a deficit in span 1 harms all 19 downstream amplifiers, but the same deficit in span 19 harms only one. Two results support this. An RBF-kernel SVM gains only 0.6 points, so the boundary really is close to linear, as the paper argued. Adding the 38 per-span residuals helps more (97.4%). **Conclusion**: two aggregated telemetry features and a linear classifier are enough to detect failing links reliably and cheaply at cloud scale, and per-span residuals then localise the fault. Future work: forecast OSNR to increase lead time, add span-position-aware features, and validate on real operator data.

References

- [1] Z. Bakhtiari, "Link Failure Prediction and Localization in Cloud Scale Networks using Supervised Learning," Stanford University. [2] S. Shalev-Shwartz et al., "Pegasos: Primal Estimated sub-GrAdient Solver for SVM," ICML 2007. [3] P. Poggiolini, "The GN model of non-linear propagation in uncompensated coherent optical systems," JLT 2012.